XRA-GS: X-ray Attenuation-Aligned Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

Sparse Tomographic View Synthesis seeks to recover unseen X-ray projections from only a few acquired views, a capability that is valuable for low-dose scanning, fast acquisition, and trajectory-constrained imaging. Existing X-ray/CT 3D Gaussian Splatting methods already employ radiative rendering, yet their Gaussian evolution still largely follows the error/gradient-driven densification rules designed for visible-light scenes. As a result, model capacity is repeatedly drawn toward high-contrast anatomical boundaries, such as bone-soft tissue interfaces, while lower-contrast interior structures that also contribute to path attenuation remain underrepresented. To address this mismatch, we present ***XRA-GS****, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis.* ***XRA-GS*** *first uses Support-Profile Seeding (SPS) to anchor initialization with the anatomical support and coarse attenuation layout provided by a coarse FDK reconstruction, then applies Geometry-aware Pruning (GAP) to remove redundant Gaussians around over-densified boundaries, and finally refines local density distributions with Adaptive Density Modulation (ADM). Under a unified protocol spanning five organs and 2/3/4-view settings,* ***XRA-GS*** *achieves the best average PSNR2D in every setting while maintaining leading or tied SSIM2D performance over prior baselines.*

1. Introduction

Novel view synthesis has recently expanded from natural-image rendering to radiographic imaging, where the goal is no longer to recover surface appearance but to predict physically valid X-ray projections from limited observations. In this paper, we study Sparse Tomographic View Synthesis: given only a few acquired X-ray views in a sparse-view setting, the model must synthesize projections at unseen angles. This problem is important for low-dose CT, fast acquisition, trajectory-constrained scanning, and cases in which patients cannot maintain long, stable scans [FDK84; AK84; WYD08]. A reliable solution would provide clinically useful supplementary views for anatomical localization, lesionrange inspection, and rapid imaging preview under severely limited sampling.

Sparse Tomographic View Synthesis differs fundamentally from natural-scene novel view synthesis. Each detector measurement reflects the remaining energy of a single X-ray after it traverses internal materials, so the observation is determined by attenuation integrated along the entire path rather than by local radiance emitted from a surface point. X-Field observed that many 3D representations were originally built around visible-light imaging, while DGR further emphasized that successful instance reconstruction depends not only on local heuristics, but on whether the representation, reconstruction target, and optimization loop are genuinely aligned with tomographic objectives [WTW*25; WLG*25]. For CT, this means that the problem should not be framed purely as surface coverage; it must also be understood as a capacity-allocation problem over the internal structures that actually participate in attenuation.

Most sparse-view 3DGS methods explain performance degradation in terms of insufficient coverage, unstable optimization, or geometric error, and therefore keep the same error/gradient-driven densification backbone while adding coverage compensation, co-regularization, or geometric/physical priors. FSGS expands coverage with proximity-guided unpooling, pseudo views, and monocular depth priors [ZFJW24]; CoR-GS stabilizes single-scene optimization through disagreement-aware co-regularization and co-pruning [ZLY*24]; DNGaussian mitigates geometric degradation through depth normalization [LZB*24]. In the X-ray/CT domain, X-Gaussian and R^2-Gaussian further align the renderer with radiative transport and projection-consistency correction [CLW*24; ZLC*24]. Yet even when alpha compositing has already been replaced by X-ray radiative projection, optimization still follows a visible-light intuition as long as primitive growth is dominated by "duplicate or split where gradients are strong." This logic prioritizes the most abrupt projection changes at anatomical boundaries, instead of the full internal path that determines attenuation.

Our central observation is therefore that the true bottleneck in extremely sparse CT is often not the number of Gaussians, but the amount of effective representation capacity assigned to attenuation-relevant structures. CT volumes are not dominated by a few visible surfaces; instead, they are occupied by continuous tissue densities. High-contrast boundaries, such as bone-soft tissue interfaces, generate sharper projection variation and thus repeatedly attract gradient-driven densification. The model consequently keeps spawning smaller and narrower Gaussians around these boundaries, while lower-contrast interior tissues behind the boundary remain weakly represented even though they also contribute to the same X-ray path. The resulting failure mode is not merely insufficient coverage, but structural redundancy and capacity misallocation.

To address this problem, we propose **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis. Rather than appending more auxiliary branches onto a conventional densification backbone, **XRA-GS** rewrites the Gaussian evolution process itself into three stages: path-anchored initialization, boundary redundancy recovery, and local density refinement. Figure 1 illustrates the key idea. The top row summarizes the standard error/gradient-driven densification pipeline commonly used in highly sparse instance reconstruction, whereas the bottom row shows how **XRA-GS** uses a coarse FDK support to anchor initialization and then reallocates capacity through

GAP and ADM. Concretely, the framework is instantiated by Support-Profile Seeding (SPS), Geometry-aware Pruning (GAP), and Adaptive Density Modulation (ADM), which respectively handle initialization, redundancy recovery, and local refinement.

Our contributions are threefold:

- We show that conventional 3DGS densification is misaligned with X-ray/CT imaging: high-gradient anatomical boundaries repeatedly attract Gaussian duplication, leading to local structural redundancy and capacity misallocation.
- We present **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework that reorganizes Sparse Tomographic View Synthesis into path-anchored initialization, boundary redundancy removal, and local density refinement, instantiated by SPS, GAP, and ADM.
- Under a five-organ protocol with 2/3/4-view sparse-view settings, **XRA-GS** achieves the best average PSNR2D in all settings and strong SSIM2D performance, with the most stable gains appearing in the 3-view regime; compared with R^2-Gaussian, the average PSNR2D improvements are +0.19/+0.26/+0.02 dB, respectively.

2. Related Work

2.1. Sparse-view Novel View Synthesis

3D Gaussian Splatting (3DGS) explicitly represents a scene with a set of anisotropic Gaussians and achieves efficient rendering through differentiable rasterization, offering a strong quality-efficiency trade-off relative to conventional NeRF formulations [KKLD23]. Since its introduction in 2023, 3DGS has been extended to structured representations, anti-aliased rendering, dynamic scenes, and compact compression [CW24; LYX*24; YCH*24; GL24; WRW*24; XLG*24; YGZ*24; NMKW24; RNSZ24]. Under sparse-view novel view synthesis, existing work has gradually split into two directions: per-scene optimization of Gaussian primitives and feed-forward prediction from a few input views.

The per-scene optimization line stays closest to the original 3DGS test-time optimization setting, but diverges in how it explains failure. FSGS mainly attributes degradation to insufficient coverage, and therefore augments densification with proximity-guided unpooling, pseudo views, and monocular depth priors [ZFJW24]. CoR-GS instead treats the problem as unstable single-scene optimization and introduces dual-field co-regularization with co-pruning to suppress geometric disagreement [ZLY*24]. DNGaussian focuses on geometric recovery errors under sparse-view inputs and explicitly injects geometric priors through global-local depth normalization [LZB*24]. More recent methods continue along structure regularization and generative pseudo-view synthesis: DropGaussian reduces training-view overfitting through random dropping and structure-aware pruning, whereas GS-GS uses diffusion-generated pseudo views to improve unseen regions and cross-view consistency [PRK25; KYW25].

The other direction is generalizable or feed-forward Gaussian prediction, which moves the burden of completing geometry and appearance priors into large-scale pretraining. pixelSplat samples Gaussian centers from probabilistic depth distributions on image pairs [CLTS24]; MVSplat further aggregates multi-view correspondence through a plane-sweeping cost volume [CXZ*24]; and DepthSplat and MonoSplat explicitly incorporate depth estimation and monocular foundation priors to improve geometric stability and cross-domain generalization [XPW*25; LFY*25]. Overall, regardless of whether the method is per-scene optimized or feed-forward, most extremely sparse NVS work still frames the core prob-

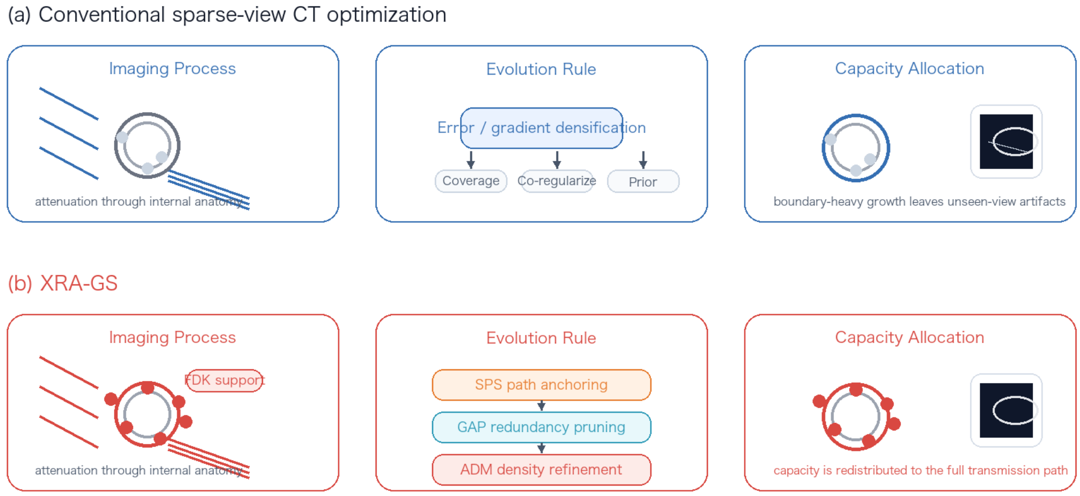

Figure 1: *Comparison between a conventional sparse-view 3DGS workflow and **XRA-GS**. The top row summarizes the common error/gradient-driven densification pipeline in highly sparse CT, which tends to place newly spawned Gaussians near high-gradient anatomical boundaries. The bottom row shows our attenuation-aligned evolution pipeline, which uses coarse FDK support for path-anchored initialization, then reclaims redundant Gaussians with GAP and refines local density distributions with ADM. The figure emphasizes that our intervention targets the Gaussian evolution rule rather than the X-ray imaging equation itself.*

lem as coverage shortage, geometric instability, or prior deficiency. In contrast, our focus is the capacity allocation problem that arises specifically in CT, where each detector pixel corresponds to attenuation along a single ray rather than multi-source radiance accumulation.

2.2. X-ray Novel View Synthesis and Tomographic Reconstruction

Conventional CT reconstruction can be broadly grouped into analytical and iterative methods. Analytical methods such as FDK rely on filtered back-projection and are computationally efficient, but they suffer from severe streak artifacts in sparse-view settings; iterative methods such as SART can incorporate prior constraints more flexibly, although at higher computational cost [FDK84; AK84]. Since 2023, learning-based sparse-view X-ray/CT methods can roughly be organized by representation into two families: implicit attenuation fields and explicit radiative primitives.

The implicit line extends neural field or neural attenuation field modeling, but with greater emphasis on X-ray imaging structure and scan geometry. Structure-Aware Sparse-View X-ray 3D Reconstruction (SAX-NeRF) jointly models 3D line dependencies and 2D projection context through a Lineformer and masked local-global ray sampling, optimizing novel view synthesis and CT reconstruction within one framework [CWY*24]. Geometry-Aware Attenuation Learning back-projects multi-view 2D features into a 3D volume-feature grid according to scan geometry and then reconstructs CBCT volumes with a 3D decoder, representing a shift from perinstance self-supervised optimization to population-level encoder-decoder inference [LFL*25]. These methods preserve the expressive power of continuous volumetric density fields, but they typically still require dense ray sampling or heavy 3D volumetric decoding.

The explicit line instead redesigns the primitive representation around X-ray penetration and attenuation. X-Gaussian introduced radiative Gaussians and differentiable X-ray rasterization for projection-domain novel view synthesis [CLW*24]. R^2-Gaussian pushed the representation further toward reconstruction consistency by correcting 3D-to-2D integration bias and introducing differentiable voxelization, allowing radiative Gaussians to better support sparse-view tomography [ZLC*24]. Building on this line, X-Field strengthens physical consistency with material-aware ellipsoids and path segmentation [WTW*25]; DGR directly reconstructs voxel grids using discretized Gaussian functions [WLG*25]; GR-Gaussian uses graph-structured gradients to suppress needle-like artifacts under extreme sparsity; and X^2-Gaussian extends radiative Gaussians to continuous-time 4D CT [YMS*25; YCZ*25]. Taken together, X-ray/CT research is moving from borrowing visible-light representations to redesigning them around transmission imaging, attenuation accumulation, and reconstruction consistency. Our work belongs to the explicit radiative-primitives line, but its contribution is not another physical branch. Instead, we argue that even with a physically aligned renderer, sparse-view CT still suffers if Gaussian capacity allocation continues to inherit the boundary-chasing logic of visible-light densification.

3. Method

Figure 1 already summarizes the core distinction at the pipeline level: **XRA-GS** does not keep stacking auxiliary branches on top of an existing densification backbone, but instead reorders how Gaussians are initialized, reclaimed, and refined. We now turn to the implementation level and show how this attenuation-aligned evolution pipeline is instantiated by SPS, GAP, and ADM on top of the radiative Gaussian backbone of R^2-Gaussian.

3.1. Problem Formulation

Let the sparse-view observation set be $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^{V}$, where I_v denotes the v-th X-ray projection and π_v denotes its corresponding acquisition geometry. Given a coarse FDK reconstruction volume V_{FDK} obtained from these observations, our goal is to learn an explicit radiative Gaussian representation in a per-case optimization setting so that unseen-angle target projections can be synthesized from only a few observed projections. Although this task is naturally coupled with volumetric density recovery, our primary optimization target remains novel-view quality in the projection domain.

3.2. 3DGS and X-ray Rendering Background

To clarify that our problem is not simply “adding another prior,” we first contrast the pixel formation mechanisms of visible-light 3DGS and X-ray projection. In standard visible-light 3DGS, the color of a pixel is typically accumulated through front-to-back compositing of multiple splats ordered along the viewing ray:

$$\mathbf{C}(\mathbf{r}) = \sum_{i=1}^{N_{\mathrm{r}}} T_i \, \alpha_i \, \mathbf{c}_i, \qquad T_i = \prod_{j<i} (1 - \alpha_j), \tag{1}$$

where $\mathbf{r}$ is the camera ray, and α_i and $\mathbf{c}_i$ denote the opacity and radiance attributes of the i-th splat. The core intuition is that a pixel is formed by the accumulation of local surface contributions.

In X-ray imaging, however, a detector pixel corresponds more closely to the remaining energy of a single ray after traversing internal materials. Its physical measurement can be written as

$$I(\mathbf{r}) = I_0 \exp\left(-\int_{\mathbf{r}} \mu(\mathbf{x})\,\mathrm{d}l\right), \qquad -\log\frac{I(\mathbf{r})}{I_0} = \int_{\mathbf{r}} \mu(\mathbf{x})\,\mathrm{d}l, \tag{2}$$

where $\mu(\mathbf{x})$ is the attenuation coefficient at spatial location $\mathbf{x}$ and $\mathrm{d}l$ is the differential path length along the ray. In other words, an X-ray pixel is not a sum of surface colors; it is the integral of attenuation contributions from different materials along a single path. What matters is how much of each material occupies the path, rather than how strong the local response is at one boundary point.

We represent the scene as a Gaussian set $\mathcal{G} = \{g_i\}_{i=1}^{N}$, where each Gaussian is written as

$$g_i = (\mathbf{p}_i, \Sigma_i, \rho_i, \mathbf{c}_i), \tag{3}$$

denoting the center, covariance, density, and appearance/radiative parameters, respectively. For any view π_v, the R^2-Gaussian-style radiative renderer can be written as

$$\hat{I}_v = \mathcal{R}(\mathcal{G}; \pi_v). \tag{4}$$

Except for our new SPS, GAP, and ADM modules, the radiative rasterization, densification, and differentiable voxelization settings follow R^2-Gaussian [ZLC*24]. Importantly, X-Gaussian and R^2-Gaussian have already aligned the renderer with X-ray radiative projection. The mismatch we target lies mainly in the Gaussian evolution rule rather than in the imaging equation itself. In this baseline, Gaussians are still usually duplicated or split according to projection-error gradients. This is effective for improving surface coverage in natural scenes, but in CT it tends to keep pushing the evolving Gaussian set toward high-gradient anatomical boundaries. The model therefore uses the right X-ray renderer, yet still organizes Gaussians with a visible-light-style growth logic, which is precisely the issue our three modules address.

3.3. Overall Framework

As shown in Figure 2, **XRA-GS** instantiates the attenuation-aligned Gaussian evolution pipeline as three sequential modules embedded into the explicit radiative Gaussian backbone of R^2-Gaussian. SPS seeds Gaussians into more relevant anatomical regions during initialization using the coarse FDK reconstruction. GAP then periodically reclaims locally redundant Gaussians after densification, preventing high-gradient boundaries from dominating Gaussian growth. Finally, ADM refines the density distribution of retained Gaussians based on position-aware context during later optimization. In short, we do not redesign X-ray physics; we reorganize the optimization loop itself into path anchoring, boundary de-redundancy, and local refinement.

3.4. Support-Profile Seeding (SPS)

SPS initializes Gaussian primitives in anatomically relevant regions using support-profile cues from the coarse FDK reconstruction. Initialization strongly influences the downstream optimization trajectory, yet neither of the standard 3DGS initializers fits CT well: uniform sampling cannot distinguish the anatomical importance of air, soft tissue, and bone, while SfM-based point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projection [CLW*24]. Crucially, we do not use a vague spatial prior. Instead, we draw on three concrete signals contained in the FDK reconstruction: foreground anatomical support, low-

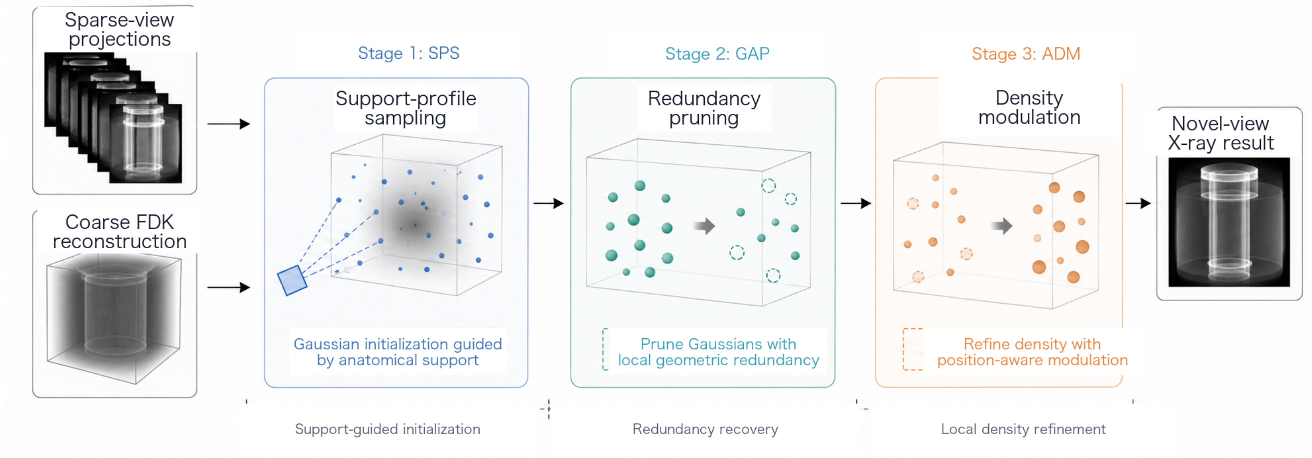

Figure 2: *Overview of **XRA-GS**. Given highly sparse CT projections and a coarse FDK reconstruction, SPS performs path-anchored initialization from foreground support and coarse attenuation layout, GAP reclaims boundary redundancy, and ADM refines density in a position-aware manner. Together they form the attenuation-aligned Gaussian evolution pipeline that produces the target novel-view X-ray projection.*

frequency attenuation layout, and a coarse intensity hierarchy that differentiates air, soft tissue, and bony structures. The sparse-view CT literature also widely treats FDK/FBP reconstruction as a coarse CT volume with streak artifacts whose main value lies in global structure layout and useful frequency content rather than reliable fine detail [WCS*22; LMC*23; LFZ*25; ZLC*24]. In bone-dominant scenes such as Head, Foot, and parts of Chest, these support profiles can even be interpreted as a coarse anatomical scaffold that remains compatible with landmark-oriented structure cues, as CBCT/CT volumes are widely used for craniofacial landmark localization and TMJ condyle-fossa morphology assessment [SBS*14; CLL*23]. Therefore, SPS is better understood as exploiting an FDK-derived, landmark-capable anatomical scaffold for initialization, while in soft-tissue-dominant scenes such as Abdomen and Pancreas the more conservative interpretation is still coarse anatomical support and attenuation layout. Based on this view, we use the coarse density volume V_{FDK} produced by FDK and define a density-weighted sampling distribution over a discrete voxel grid Ω:

$$p(\mathbf{x}) = \frac{(\max(V_{\text{FDK}}(\mathbf{x}), \epsilon))^{\gamma}}{\sum_{\mathbf{x}'\in\Omega} (\max(V_{\text{FDK}}(\mathbf{x}'), \epsilon))^{\gamma}}, \tag{5}$$

where ϵ avoids zero probability in background regions and γ controls sampling concentration.

Density weighting alone would over-concentrate Gaussians near the brightest bony structures and could mistake the strongest sparse-view FDK responses for precise geometry. We therefore mix uniform sampling with the support-profile distribution:

$$q(\mathbf{x}) = \alpha\frac{1}{|\Omega|} + (1-\alpha)p(\mathbf{x}), \quad \mathbf{x}\in\Omega, \tag{6}$$

and independently sample M initial centers $\{\mathbf{x}_m\}_{m=1}^{M}$ from $q(\mathbf{x})$, where M is the number of initial Gaussians. This design preserves global coverage while allocating more Gaussians to anatomical regions with meaningful attenuation support, without collapsing

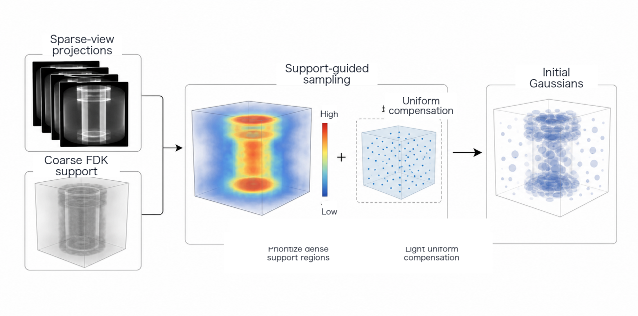

Figure 3: *Illustration of SPS. The coarse FDK reconstruction provides foreground support and coarse attenuation layout to guide density-weighted sampling, so that more initial Gaussians fall into anatomically relevant regions while a small amount of uniform compensation sampling preserves global coverage.*

entirely onto dense bony boundaries. After sampling the locations, we initialize Gaussian densities from the FDK values and set scale parameters according to the local sampling density, yielding 50K initial Gaussians. In all experiments, α and γ are selected by a small local search and then fixed to avoid scene-specific manual tuning.

3.5. Geometry-aware Pruning (GAP)

GAP periodically removes locally redundant Gaussians based on spatial proximity and gradient activity. In CT, gradient-driven densification often keeps duplicating Gaussians around bone-soft tissue boundaries because these regions produce larger projection-error gradients. However, high gradient does not always imply missing coverage; it may also indicate that the same anatomical boundary is being represented repeatedly by neighboring Gaussians. Instead of following the FSGS logic of adding support points in sparse regions [ZFJW24], GAP explicitly measures local crowding in world coordinates and uses it for redundancy recovery in over-densified regions.

Specifically, for each Gaussian G_i, let $\mathcal{N}_K(i)$ denote the index set of its K nearest neighbors excluding itself. We compute the mean Euclidean distance to these neighbors as a proximity score:

$$s_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2. \tag{7}$$

When s_i falls below a threshold τ, the Gaussian lies in an overly crowded region. To avoid deleting Gaussians that are still learning useful structure, we further introduce a gradient-activity filter. Let $\bar{g}_i$ denote the accumulated L_1 statistic of the positional gradient of this Gaussian over the current densification cycle. We define a retention mask $m_i \in \{0, 1\}$ as

$$m_i = \begin{cases} 1, & s_i \geq \tau \ \vee \ \bar{g}_i \geq \delta, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

where $m_i = 0$ means that the Gaussian is treated as a redundancy candidate in the current cycle. In implementation, GAP does not replace the original 3DGS densification; it is a periodic recovery step applied afterwards. We first generate candidate Gaussians through the standard duplication/splitting rule and then use m_i to filter redundant ones. In each cycle, we prune at most a fraction β_{prune} of the total Gaussians, and the module is triggered periodically from 2K to 20K iterations. To avoid suddenly making local support too sparse after pruning, we only shrink the covariance of newly retained Gaussians:

$$\Sigma_i^{\text{new}} = \eta \Sigma_i, \quad 0 < \eta < 1. \tag{9}$$

GAP therefore regularizes the densification process by suppressing locally redundant Gaussians, preventing high-gradient boundaries from continuously monopolizing representation capacity.

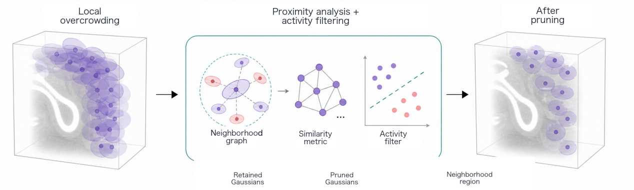

Figure 4: *Illustration of GAP. We first identify locally over-crowded regions through world-coordinate proximity, then combine this measure with gradient activity to delete only redundant Gaussians with limited remaining learning value.*

3.6. Adaptive Density Modulation (ADM)

ADM predicts position-dependent density offsets to refine the density distribution of retained Gaussians. Even after better initialization and redundancy control, density updates remain highly heterogeneous across anatomy: bone usually corresponds to higher attenuation, air regions in the lung should remain low-density, and soft tissue often requires smoother transitions. If all Gaussians share the same point-wise gradient update rule, the model struggles to express this spatial heterogeneity. We therefore encode Gaussian centers with K-Planes and use a lightweight decoder that predicts only density-related offsets and confidence gates, without rewriting Gaussian centers, covariances, or other attributes. The role of ADM is local density refinement rather than serving as the sole source of performance.

It is important to emphasize that, although ADM contains an MLP, it is not an offline pre-trained prediction head and does not rely on extra density supervision. Both the K-Planes features and the dual-head MLP parameters are updated only during optimization on the current case, and they are trained end-to-end through the main rendering loss. In this sense, ADM is an online joint-optimization branch that adaptively amplifies or suppresses density responses in different regions while the main radiative Gaussian backbone is being trained.

Given a position $\mathbf{x}$, we first project it onto normalized 2D coordinates $\Pi_{uv}(\mathbf{x})$ on three orthogonal planes and then bilinearly sample the corresponding feature maps:

$$f_{uv}(\mathbf{x}) = \mathcal{B}(P_{uv}, \Pi_{uv}(\mathbf{x})), \quad uv \in \{xy, xz, yz\}, \tag{10}$$

$$F(\mathbf{x}) = \mathrm{concat}\big(f_{xy}(\mathbf{x}), f_{xz}(\mathbf{x}), f_{yz}(\mathbf{x})\big), \tag{11}$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling, P_{xy}, P_{xz}, and P_{yz} are the three orthogonal feature planes, and $\mathrm{concat}(\cdot)$ denotes channel-wise concatenation. K-Planes encoding was originally introduced for explicit spatiotemporal radiance fields and provides continuous spatial context at low cost [FYT*23]. Based on this encoding, a dual-head MLP predicts a density offset $o(\mathbf{x})$ and a confidence gate $c(\mathbf{x})$:

$$\big(o(\mathbf{x}), c(\mathbf{x})\big) = h(F(\mathbf{x})). \tag{12}$$

We then center the offsets by the batch mean and apply a view-dependent scaling:

$$\tilde{o}(\mathbf{x}) = \beta(t)\, s_{\text{view}}\, r_{\max}\, c(\mathbf{x}) \cdot \big(o(\mathbf{x}) - \bar{o}\big), \tag{13}$$

where $\bar{o}$ is the mean offset of the current batch, $r_{\max}$ is the maximum modulation amplitude, and s_{view} is a scaling factor that varies with the number of input views. To prevent modulation from disrupting basic geometric convergence early in training, we use a warm-up/hold/decay schedule for $\beta(t)$:

$$\beta(t) = \begin{cases} \frac{t}{t_w}, & 0 \le t < t_w, \\ 1, & t_w \le t < t_h, \\ \exp\big(-\frac{t-t_h}{\tau}\big), & t \ge t_h. \end{cases} \tag{14}$$

The final density is

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \big(1 + \tilde{o}(\mathbf{x})\big). \tag{15}$$

Therefore, ADM learns not a global density boost,

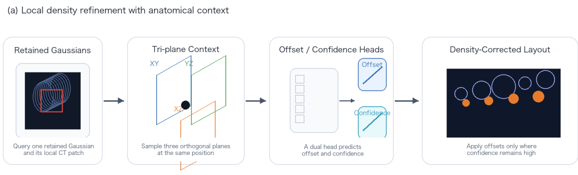

Figure 5: *Illustration of ADM. K-Planes provide continuous spatial context, and a dual-head predictor estimates position-dependent density offsets and confidence gates to realize anatomy-aware density modulation.*

but a spatial-context-aware relative correction that better matches the locally continuous yet regionally heterogeneous structure of CT data.

3.7. Training Loss

The three modules of **XRA-GS** and the radiative rendering backbone of R^2-Gaussian form a single end-to-end training pipeline. During training, we supervise reprojection results on the observed views with the following overall loss:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_{\text{dssim}}\mathcal{L}_{\text{dssim}} + \lambda_{\text{3DTV}}\mathcal{L}_{\text{3DTV}} + \lambda_{\text{pTV}}\mathcal{L}_{\text{pTV}}, \tag{16}$$

where

$$\mathcal{L}_1 = \frac{1}{V}\sum_{v=1}^{V}\|I_v - \hat{I}_v\|_1, \tag{17}$$

$$\mathcal{L}_{\text{dssim}} = \frac{1}{V}\sum_{v=1}^{V}\text{DSSIM}(I_v, \hat{I}_v), \tag{18}$$

$$\mathcal{L}_{\text{3DTV}} = \sum_{i=1}^{N}\sum_{d\in\{x,y,z\}}|\nabla_d \rho_i|\,, \tag{19}$$

$$\mathcal{L}_{\text{pTV}} = \sum_{p\in\{xy,xz,yz\}}\omega_p(t)\,\text{TV}(P_p). \tag{20}$$

Here, $\omega_p(t)$ is the time-varying regularization weight for the p-th feature plane. $\mathcal{L}_1$ and $\mathcal{L}_{\text{dssim}}$ jointly constrain projection reconstruction quality, $\mathcal{L}_{\text{3DTV}}$ encourages spatial smoothness in volumetric density, and $\mathcal{L}_{\text{pTV}}$ imposes planar total-variation regularization on the K-Planes features. During training, the plane features and MLP parameters of ADM are jointly updated with the backbone Gaussian parameters under the same loss, so density modulation is learned online rather than fixed in advance. Since SPS only changes initialization, GAP is triggered periodically during training, and ADM applies lightweight density modulation, the basic X-ray rendering process at inference remains unchanged and the extra cost is concentrated in training.

4. Experiments

4.1. Experimental Setup

Data and task setting. We use the processed datasets released in the official X-Gaussian repository. Among the five scenes used in prior work, Chest comes from LIDC-IDRI, while Head, Abdomen, Foot, and Pancreas come from the open scientific visualization dataset. The package follows the projection generation and preprocessing protocol of the NAF/X-Gaussian line and organizes the data into a unified format. On top of this public data, we further construct a sparse-view setting by keeping only 2/3/4 input projections for each case. All models are trained only on these observed projections and evaluated on unseen angles in terms of novel-view projection quality.

Baselines. Our main comparisons include CoR-GS [ZLY*24], DNGaussian [LZB*24], FSGS [ZFJW24], R^2-Gaussian [ZLC*24], X-Gaussian [CLW*24], and same-protocol X-Field [WTW*25]. X-Gaussian, R^2-Gaussian, and X-Field serve as the most direct X-ray/CT baselines, representing radiative Gaussian novel view synthesis, reconstruction consistency, and material-aware direct modeling, respectively. CoR-GS, DNGaussian, and FSGS are representative sparse-view 3DGS references from natural-scene NVS and help reveal how different densification workflows fail under tomographic projection.

Metrics. We use projection-domain PSNR2D and SSIM2D as the primary metrics. Following the official X-Gaussian evaluation script, rendered projections and ground-truth projections are first normalized to $[0, 1]$, then each test projection is evaluated independently and averaged over all test views of the same scene. PSNR2D is computed as $20 \log_{10}(1/\sqrt{\text{MSE}})$, where MSE is the mean squared error over the full projection. SSIM2D is computed with an 11×11 Gaussian window, $\sigma = 1.5$, and the standard constants $C_1 = 0.01^2$ and $C_2 = 0.03^2$.

Implementation details. SPS uses $\alpha = 0.2$, $\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ neighbors, threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\text{prune}} = 0.02$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64 with feature dimension 32 and is activated after 15K iterations. All models are trained for 30K iterations.

4.2. Main Results

Tables 1 and 2 report the main PSNR2D and SSIM2D results, respectively. For PSNR2D, we report per-organ values for CoR-GS, DNGaussian, FSGS, X-Gaussian, R^2-Gaussian, and **XRA-GS**, and additionally include the same-protocol average of X-Field. For SSIM2D, only the averages of X-Field, R^2-Gaussian, and **XRA-GS**are available in the source logs, so we leave the unavailable same-protocol cells blank.

In PSNR2D, **XRA-GS** achieves the best average result at all three sparsity levels, with the most stable gains concentrated in the 3-view regime. Compared with R^2-Gaussian, the average PSNR2D gains of **XRA-GS** are +0.19/+0.26/+0.02 dB under 2/3/4-view settings, respectively. Compared with X-Field, the corresponding average gains are +0.95/+3.12/+2.79 dB. These improvements support our main claim: when the input information is still too limited for stable reconstruction, path anchoring and redundancy recovery allocate capacity more effectively to the interior structures that actually govern attenuation.

The available SSIM2D results show the same trend as PSNR2D: **XRA-GS** achieves the best average SSIM2D under 2-view and 3-view settings, and ties R^2-Gaussian at the third decimal place in 4-view. Taken together, both metrics indicate that the strongest operating regime of our method is the moderately sparse 3-view setting.

Figure 6 presents representative qualitative comparisons. Keeping both natural-scene sparse-view 3DGS references and direct X-ray baselines, it reveals two distinct error patterns: the former tend to exhibit blur, streaking, and boundary leakage, whereas the latter better preserve the overall anatomical attenuation pattern. On top of that, **XRA-GS** further suppresses background noise, boundary diffusion, and local spillover in scenes such as Chest and Pancreas.

To further isolate local differences, Figure 7 enlarges two sensitive regions corresponding to the lower-chest boundary in Chest 2-view and a local structure in Foot 2-view. Compared with FSGS and CoR-GS, **XRA-GS** clearly reduces streak-like spread and local over-bright accumulation around these high-gradient regions; compared with R^2-Gaussian, it also yields tighter boundary behavior with less blurry spillover.

Cross-view consistency. Beyond per-organ averages, we also examine stability across unseen angles. Figure 8 shows a Chest 2-view case at three repre-

Table 1: *Main PSNR2D comparison under the unified five-organ, 2/3/4-view protocol. Blank entries denote same-protocol results that are unavailable in the source logs, so we do not estimate or extrapolate them. Best and second-best results are highlighted in bold and underline, respectively.*

Method	Chest			Head			Abdomen		
	2v	**3v**	**4v**	**2v**	**3v**	**4v**	**2v**	**3v**	**4v**
DNGaussian	14.30	16.02	15.76	15.72	18.04	16.24	19.33	23.51	24.72
CoR-GS	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03
FSGS	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63
X-Gaussian	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85
R^2-Gaussian	<u>21.05</u>	<u>26.22</u>	<u>25.47</u>	<u>23.54</u>	<u>26.60</u>	**28.66**	<u>24.85</u>	<u>29.26</u>	<u>30.56</u>
X-Field*									
XRA-GS	**21.26**	**26.81**	**25.83**	**23.63**	**26.63**	<u>28.52</u>	**24.97**	**29.73**	**30.89**

Method	Foot			Pancreas			Avg		
	2v	**3v**	**4v**	**2v**	**3v**	**4v**	**2v**	**3v**	**4v**
DNGaussian	20.17	23.51	19.33	16.15	22.43	25.47	17.13	20.70	20.30
CoR-GS	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
FSGS	<u>22.17</u>	25.37	27.08	<u>23.03</u>	25.27	26.22	20.25	23.10	24.57
X-Gaussian	**22.34**	25.48	26.92	**23.21**	25.12	25.55	20.38	23.19	24.34
R^2-Gaussian	19.39	<u>28.48</u>	**30.04**	17.83	<u>28.61</u>	**30.90**	<u>21.33</u>	<u>27.83</u>	<u>29.13</u>
X-Field*							20.57	24.97	26.41
XRA-GS	19.55	**28.49**	<u>29.86</u>	17.80	**29.43**	**30.90**	**21.44**	**28.22**	**29.20**

sentative novel views, 0°, 45°, and 90°. The results indicate that the advantage of **XRA-GS** over R^2-Gaussian is not restricted to one particular angle, but remains visible across multiple unseen views through a more stable global attenuation pattern.

Comparison with same-protocol X-Field. The X-Field* row in Table 1 shows that the gains of **XRA-GS** do not rely on natural-scene sparse-view baselines. Even against a material-aware direct X-ray/CT model that is closer to our setting, the attenuation-aligned evolution strategy still retains a clear average advantage.

Applicability boundary. These averages also define the operating boundary of the method. In the extremely under-determined 2-view setting, the coarse support and attenuation layout from FDK still cannot fully replace missing projection evidence. At 4 views, the reconstruction-consistency advantage of

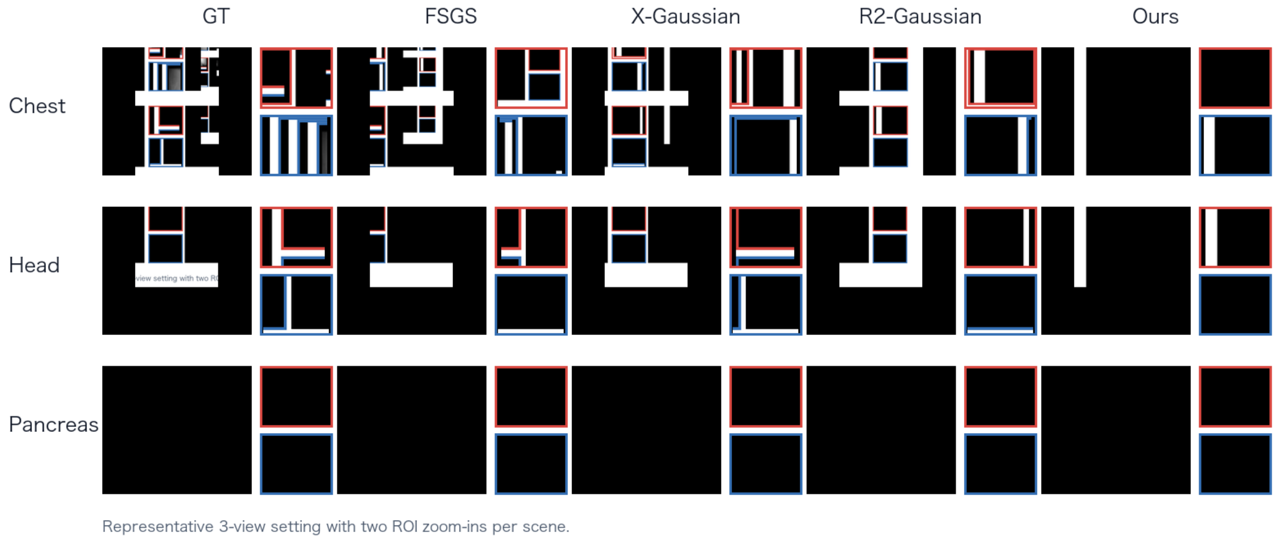

Figure 6: *Main qualitative comparison. Each row corresponds to a representative organ scene. The columns show GT, FSGS, X-Gaussian, R^2-Gaussian, and* ***XRA-GS****, together with two local zoom-ins. Compared with natural-scene densification references, direct X-ray baselines better preserve the global anatomical attenuation layout; on top of that,* ***XRA-GS*** *further suppresses background noise, boundary leakage, and local artifact spread.*

R^2-Gaussian becomes competitive again and can approach or match our result in some cases. Accordingly, the main strength of **XRA-GS** lies in the moderately sparse 3-view regime, where enough information remains for capacity redistribution to matter.

4.3. Module and Initialization Analysis

Table 3 first isolates the accuracy contribution of the three modules. Because the currently archived experiment assets only preserve this set of average PSNR2D results, we use the table strictly to answer which module delivers the main gain, rather than extrapolating to training efficiency, memory, or Gaussian count. The results show that GAP remains the dominant contributor: adding GAP alone improves the five-organ 3-view average PSNR2D from 27.80 dB to 28.22 dB and reaches the same average upper bound as the full model in 4-view. By contrast, SPS behaves more like an initialization gain under low-view settings, while the standalone effect of ADM is smaller and more consistent with a late-stage refinement module.

Figure 10 provides the local visual evidence corresponding to Table 3. We select two regions that most clearly expose boundary bias and background spillover, showing that the main gain from GAP is visible in local residual structure rather than only in average PSNR2D.

Table 4 isolates the initialization strategy. Compared with random initialization, SPS-guided seeding improves average PSNR2D at all three sparsity levels, including an increase from 27.80 dB to 28.01 dB in 3-view. This result shows that placing Gaus-

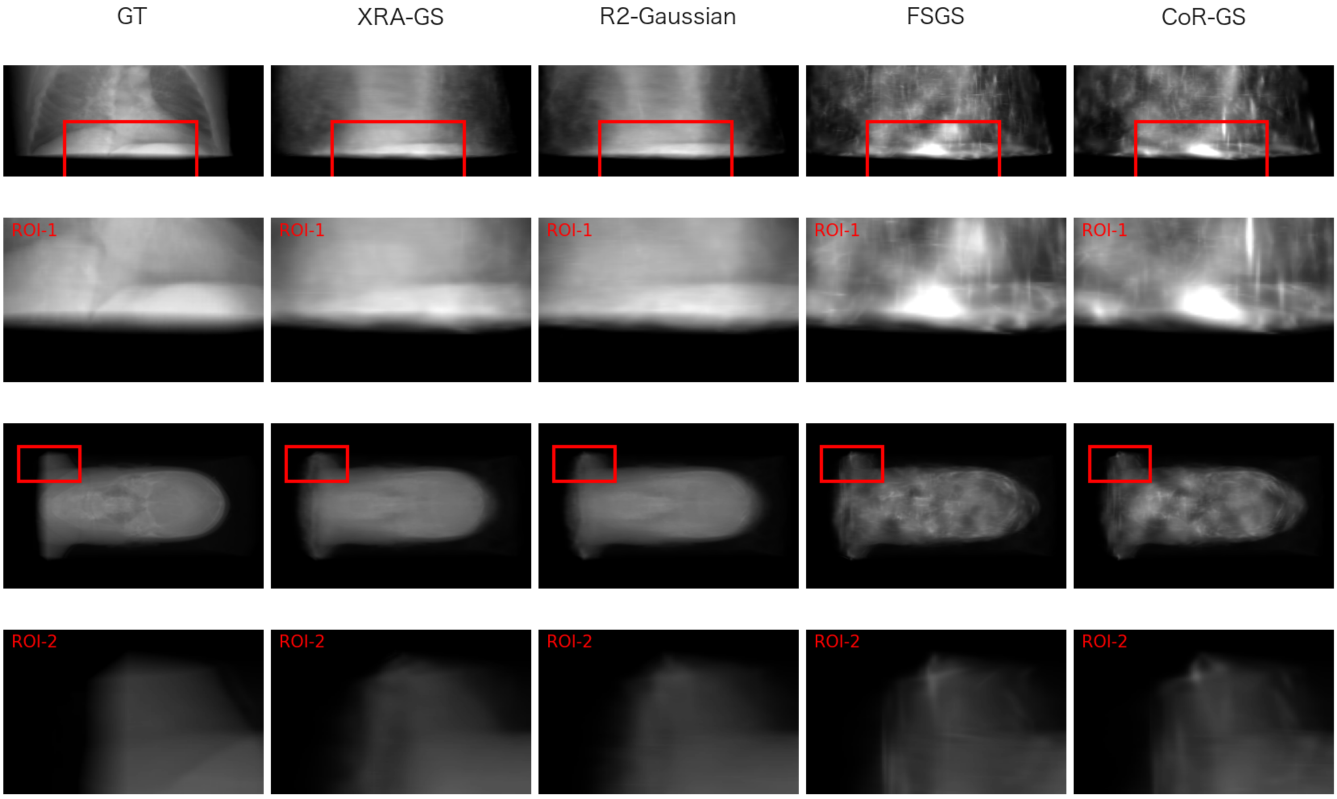

Figure 7: *Local zoom-in qualitative results. The two ROIs come from representative Chest 2-view and Foot 2-view cases. Compared with FSGS and CoR-GS, **XRA-GS** exhibits fewer streak artifacts and less local over-exposure near high-gradient boundaries; compared with R^2-Gaussian, it further reduces boundary spillover and background diffusion.*

sians first into more relevant attenuation-support regions is already useful. The full **XRA-GS** then builds on this foundation with GAP and ADM for capacity recovery and local refinement.

Figure 11 further visualizes this process as an evolution trajectory of Gaussian distribution. Compared with uniform initialization, SPS first pushes Gaussians toward more relevant foreground support regions. GAP then reclaims obviously redundant clusters near the boundary. The final **XRA-GS** representation retains the main attenuation contour while producing a more compact capacity allocation. The goal of this figure is to support our mechanism explanation of capacity misallocation and recovery, rather than to serve as a standalone accuracy claim.

4.4. View-count Analysis

Table 5 summarizes the input-view analysis under our unified protocol. As the number of inputs increases from 2 to 4, the average PSNR2D of **XRA-GS** improves from 21.52 dB to 29.15 dB and the average SSIM2D improves from 0.797 to 0.924. This confirms that additional views continuously improve overall projection reconstruction quality. However, from the perspective of our central argument, the 3-view regime remains the most informative operating

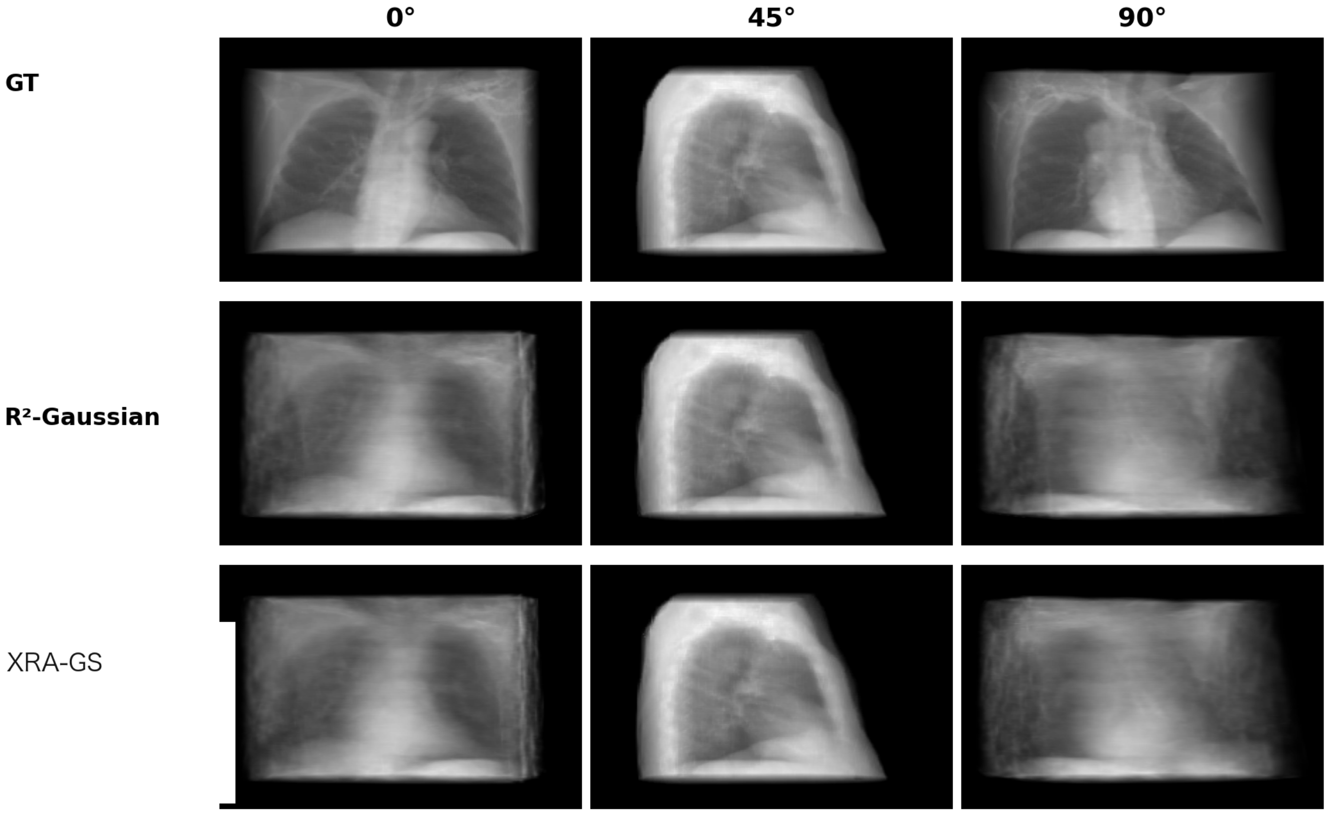

Figure 8: *Cross-view consistency visualization for a Chest 2-view case. The three columns correspond to novel views at 0°, 45°, and 90°. Compared with R^2-Gaussian,* ***XRA-GS*** *preserves a more stable global attenuation contour and less local error spread across multiple unseen angles.*

point because it is where capacity misallocation and redundancy control are most directly observable.

4.5. Hyperparameter Sensitivity

We further report hyperparameter-sensitivity results from the final sweep. Figure 12 summarizes local sweeps of four key hyperparameters in the 3-view setting: the SPS sampling mixture coefficient α, the GAP proximity threshold τ, the maximum pruning ratio β_{prune}, and the ADM warm-up start iteration. The current evidence supports a relatively concentrated working point at $\alpha = 0.2$, $\tau = 0.015$, $\beta_{\text{prune}} = 2\%$, and ADM activated after 15K iterations. We use these results as a sensitivity analysis only, rather than claiming broad robustness over a wide range.

For GAP specifically, we also keep a more focused local sweep in Figure 13. It separately visualizes the trends for the proximity threshold τ and the maximum pruning ratio β_{prune}. The best result remains close to the operating point used in the paper, namely $\tau = 0.015$ and $\beta_{\text{prune}} = 2\%$. When pruning becomes too aggressive or the threshold deviates from this point, the average PSNR2D drops noticeably.

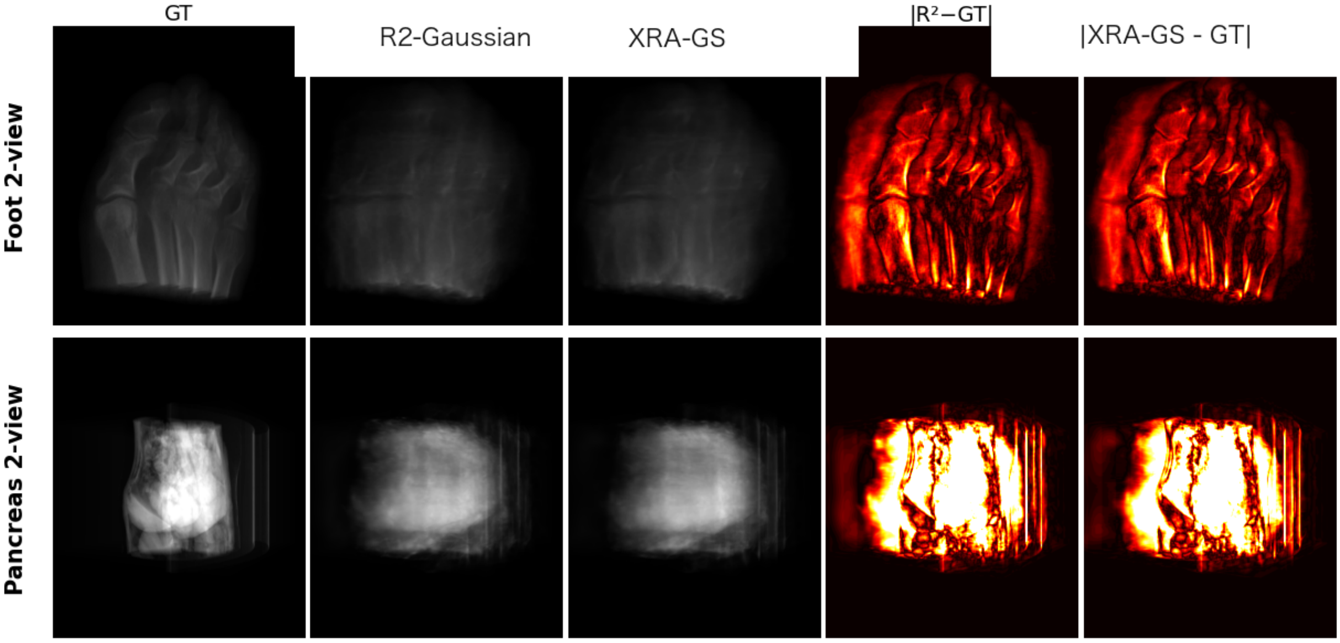

Figure 9: *Representative failure cases. We show two more extreme under-determined cases, Foot 2-view and Pancreas 2-view, together with error maps of R^2-Gaussian and **XRA-GS** relative to GT. When the input information is too limited or local structures are highly ambiguous, our method can still reduce error spread in some regions but cannot fully recover fine-grained detail.*

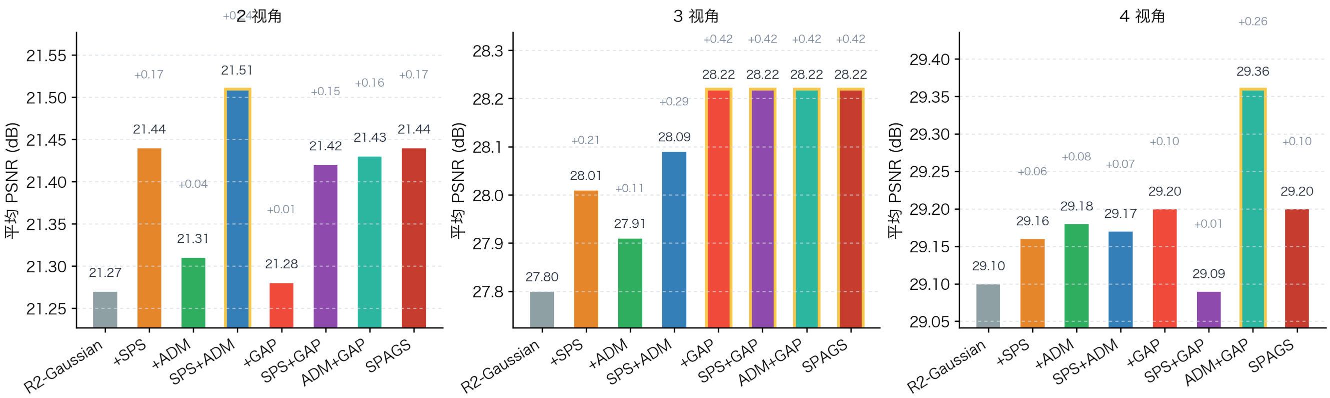

Figure 10: *Representative qualitative module ablation. Each row shows a region that is prone to boundary bias or background leakage; the top row gives predictions and the bottom row shows the corresponding residual maps* $|\mathrm{pred} - \mathrm{gt}|$. *Compared with the baseline, adding GAP alone already suppresses local over-allocation near the boundary and background leakage, while the full **XRA-GS** further tightens residual spread.*

Table 2: *Main SSIM2D comparison under the same protocol. Blank entries indicate same-protocol results that are unavailable in the source logs, so we report only the available values. Best and second-best results are highlighted in bold and underline, respectively.*

Method	**Chest**			**Head**			**Abdomen**		
	2v	**3v**	**4v**	**2v**	**3v**	**4v**	**2v**	**3v**	**4v**
DNGaussian									
CoR-GS									
FSGS									
X-Gaussian									
R^2-Gaussian									
X-Field*									
XRA-GS									

Method	**Foot**			**Pancreas**			**Avg**		
	2v	**3v**	**4v**	**2v**	**3v**	**4v**	**2v**	**3v**	**4v**
DNGaussian									
CoR-GS									
FSGS									
X-Gaussian									
R^2-Gaussian							0.794	0.903	**0.924**
X-Field*							0.717	0.850	0.880
XRA-GS							**0.797**	**0.904**	**0.924**

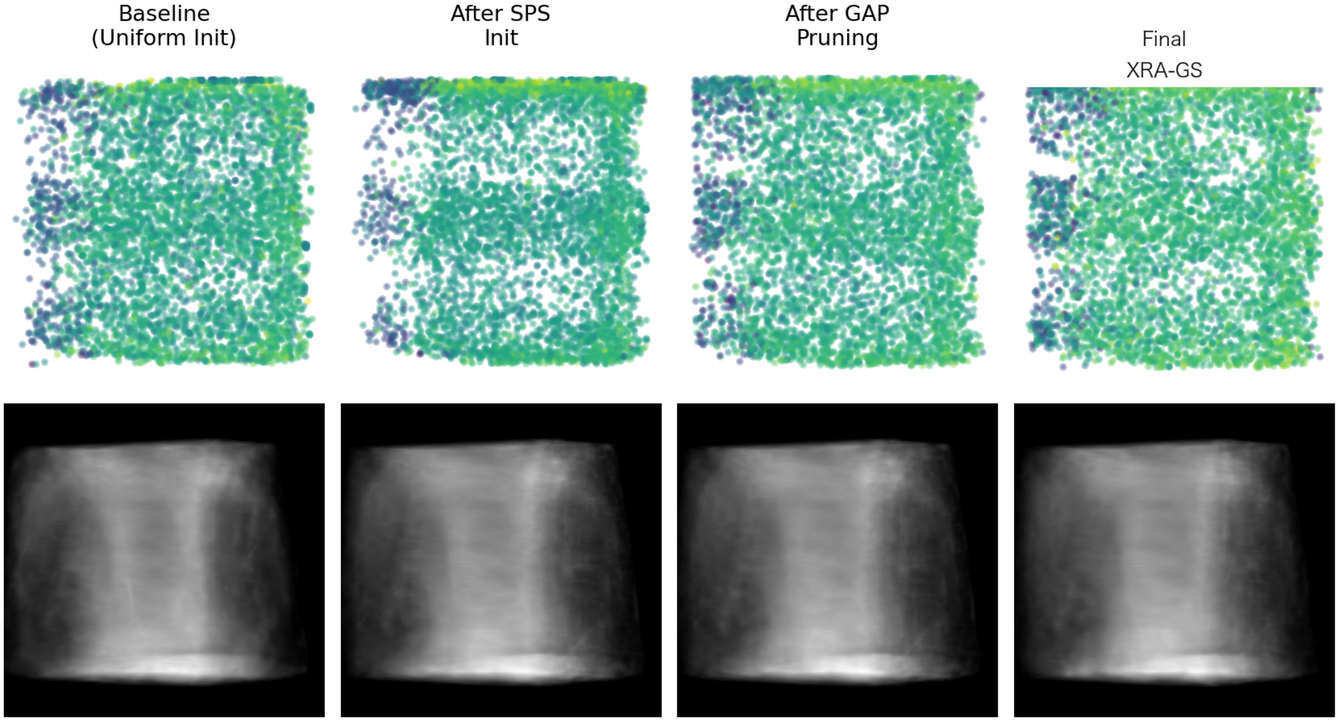

Figure 11: *Visualization of Gaussian spatial distribution evolution. From left to right we show the baseline with uniform initialization, SPS initialization, the result after GAP recovery, and the final* ***XRA-GS*** *representation. The top row shows the Gaussian distribution in a representative slice, and the bottom row shows the corresponding projection results, illustrating how capacity gradually shifts from indiscriminate coverage to effective attenuation-support regions.*

Table 3: *Incremental component ablation. We report the five-organ average PSNR2D under the unified 2/3/4-view protocol to isolate accuracy-side component contributions, rather than to support claims about training efficiency or memory usage. GAP provides the clearest primary gain, SPS mainly improves low-view initialization, and ADM behaves more like a late-stage refinement module.*

Method	2-view Avg.	3-view Avg.	4-view Avg.
Baseline	21.27	27.80	29.10
+SPS	**21.44**	28.01	29.16
+ADM	21.31	27.91	29.18
+GAP	21.28	**28.22**	**29.20**
XRA-GS	**21.44**	**28.22**	**29.20**

Table 4: *Initialization comparison. We use random initialization without SPS/GAP/ADM as the baseline, and compare it with the SPS-only initialization variant and the full **XRA-GS** pipeline. SPS-guided seeding already improves low-view performance consistently, while the full pipeline further boosts 3-view and 4-view quality.*

Initialization	2-view Avg.	3-view Avg.	4-view Avg.
Random	21.27	27.80	29.10
SPS-guided	**21.44**	28.01	29.16
XRA-GS	**21.44**	**28.22**	**29.20**

Table 5: *Input-view analysis. We report the average PSNR2D and SSIM2D of **XRA-GS** under the unified protocol. Reconstruction quality improves steadily as the number of input views increases from 2 to 4, while the 3-view regime remains the clearest operating point for observing the benefit of capacity reallocation.*

#Views	PSNR2D↑	SSIM2D↑
2	21.52	0.797
3	28.09	0.904
4	**29.15**	**0.924**

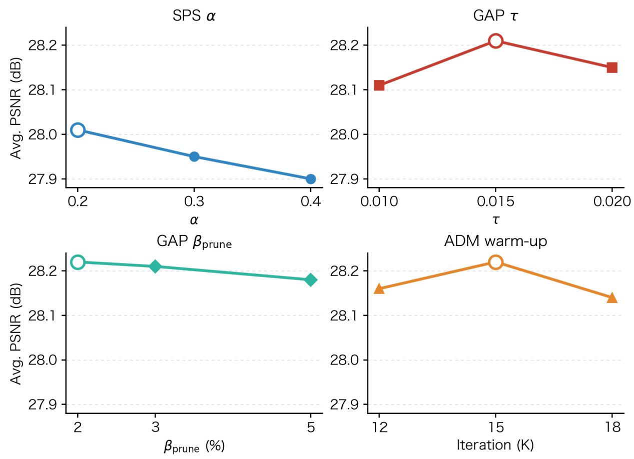

Figure 12: *Hyperparameter sensitivity analysis in the 3-view setting. The four subplots correspond to the SPS sampling mixture coefficient α, the GAP proximity threshold τ, the maximum pruning ratio β_{prune}, and the ADM warm-up start iteration. The available results support a concentrated operating point rather than broad parameter insensitivity.*

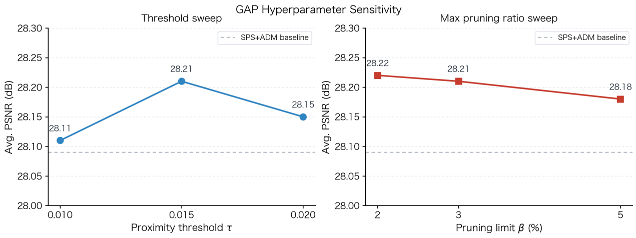

Figure 13: *Local hyperparameter sweep for GAP. The left panel studies the proximity threshold τ, and the right panel studies the maximum pruning ratio β_{prune}. The sweep suggests that the operating point used in the paper lies close to a local optimum in both curves.*

5. Conclusion

We presented **XRA-GS**, an X-ray Attenuation-Aligned Gaussian Splatting framework for Sparse Tomographic View Synthesis. Our main claim is that, under transmissive X-ray imaging, the primary bottleneck in highly sparse reconstruction is often not the lack of Gaussian primitives, but the fact that conventional densification still follows a visible-light-style boundary-chasing logic. This causes representation capacity to concentrate around high-contrast boundaries instead of adequately modeling the internal material attenuation along each ray path. Based on this diagnosis, SPS, GAP, and ADM jointly form an attenuation-aligned Gaussian evolution pipeline through path-anchored initialization, boundary de-redundancy, and local density refinement. Results on five organs, comparison against same-protocol X-Field, and analyses of modules, initialization, and view count all indicate that this evolution strategy is most effective in the 3-view regime, where its gain comes mainly from better capacity allocation rather than from simply increasing the number of primitives. In future work, we plan to explore stronger organ-adaptive priors and extend the framework to cone-beam and dynamic CT settings.

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